

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, technology adoption, disruptions, and security trends

Audience: Technology executives and network infrastructure leaders

Source: Cloudflare Radar Year in Review 2024

17% Global Traffic Growth and Starlink's 3.3x Surge

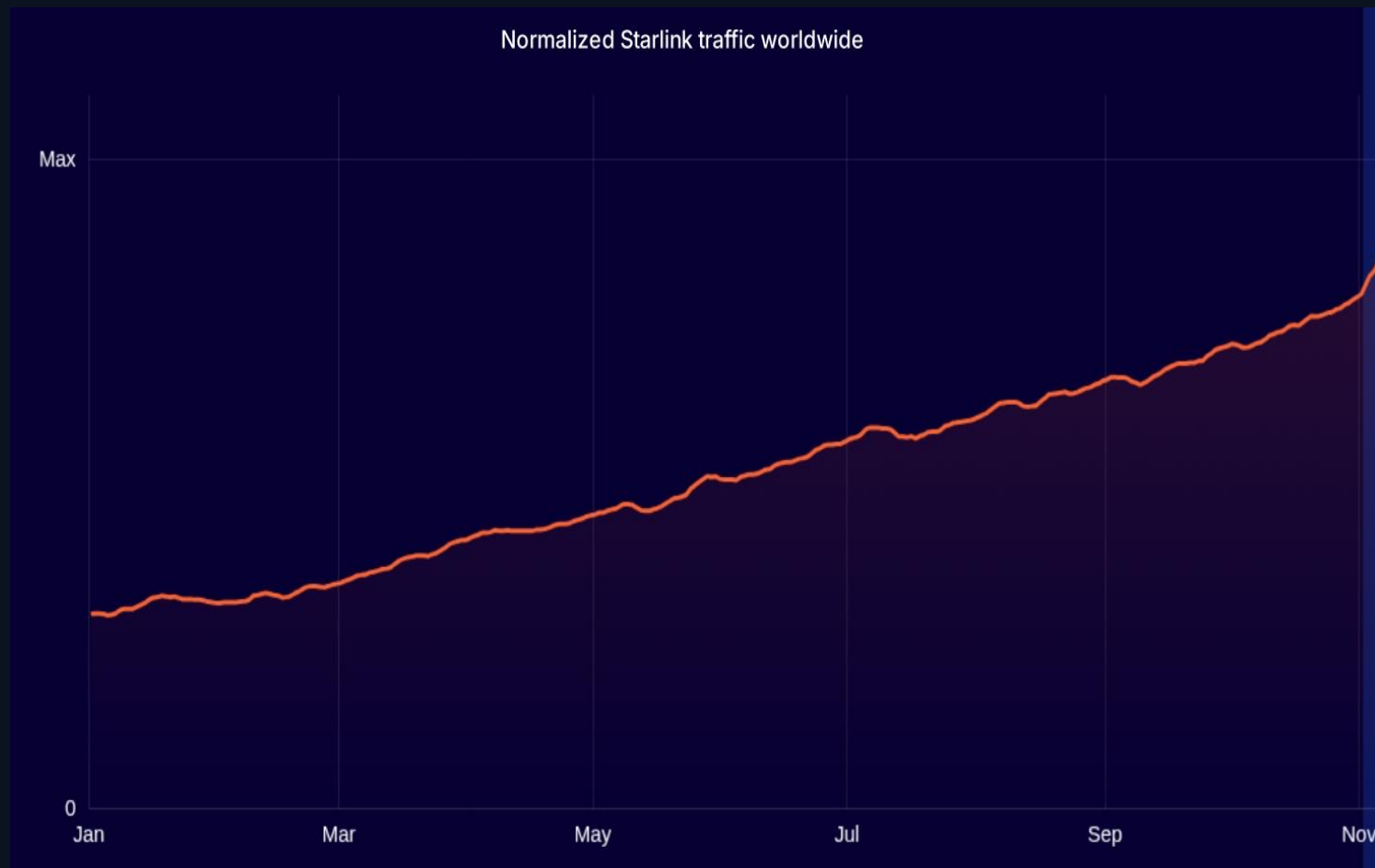
Global traffic growth accelerates as satellite connectivity scales

17%

Global Internet traffic
YoY growth in 2024

3.3x

SpaceX Starlink traffic
growth in 2024



Starlink growth reflects expanded access in previously underserved locations.

Google Leads Top 10 Services as Generative AI Rises

Cloudflare 1.1.1.1 DNS aggregate rankings for 2024

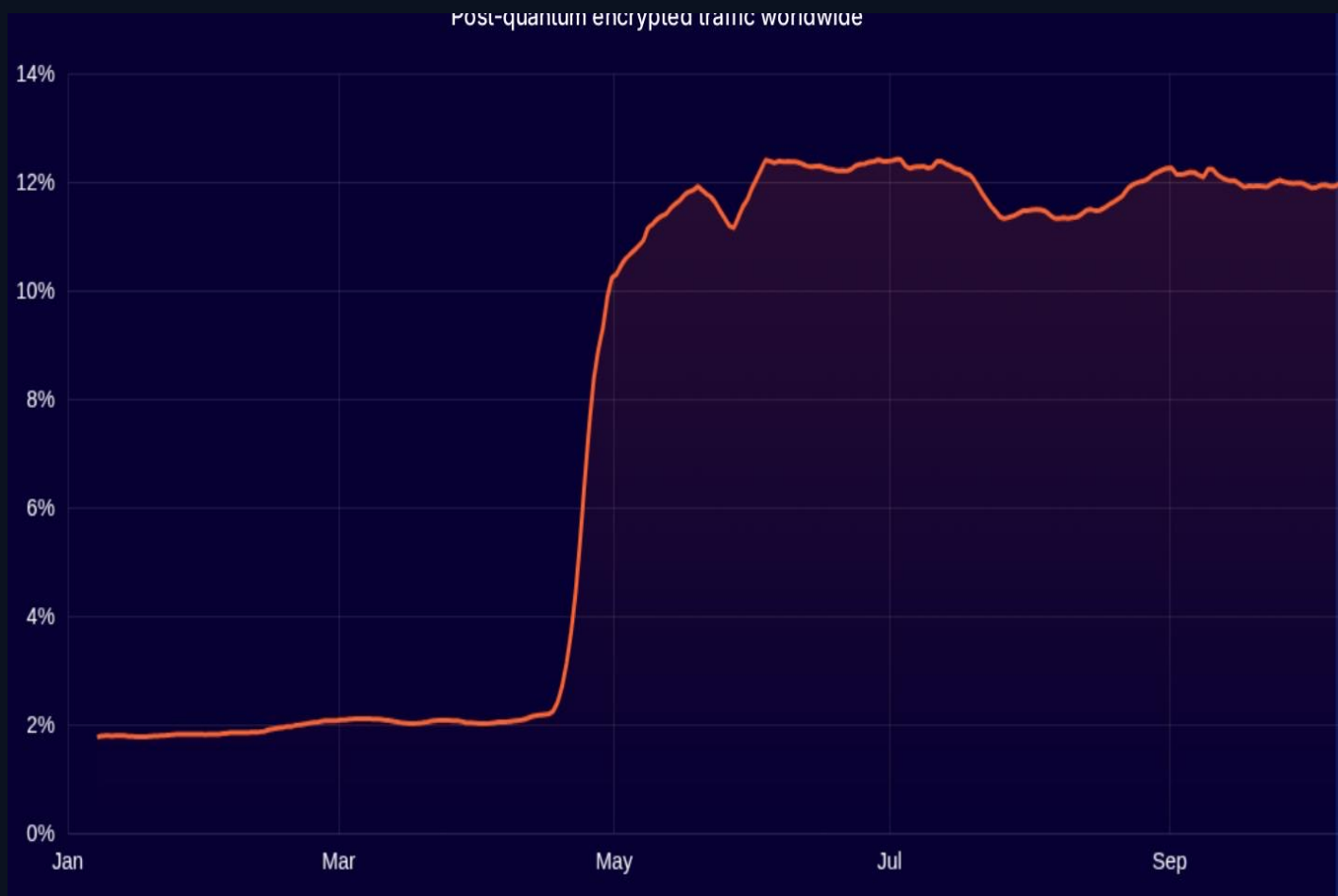
Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

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#10	WhatsApp

Generative AI continued its meteoric rise throughout 2024, while the metaverse receded.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Encryption, protocol, and developer ecosystem shifts



2% → 13%

Post-quantum share of TLS 1.3 traffic after Chrome 124 default enablement (Apr 17, 2024).

21%

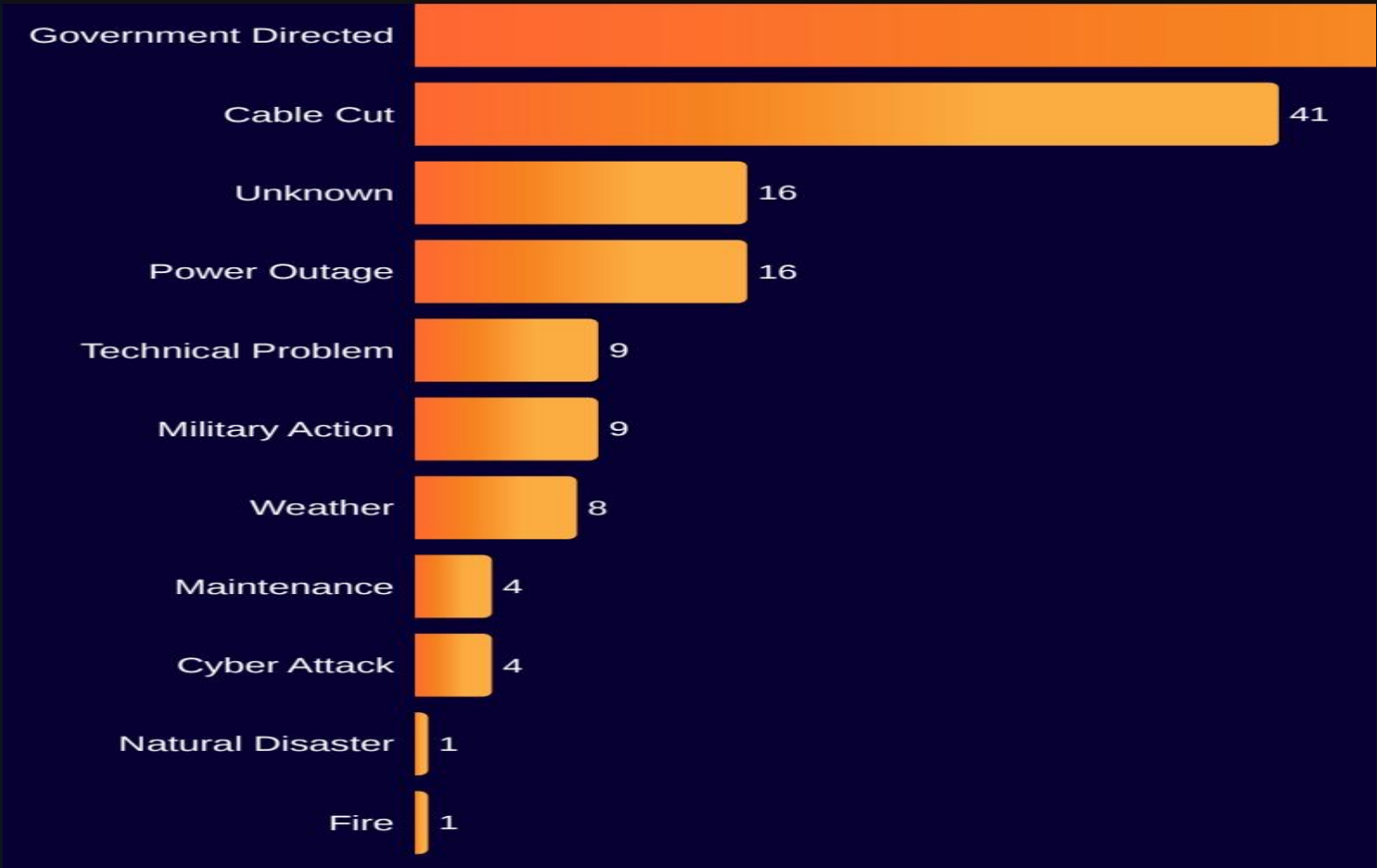
of global traffic uses HTTP/3 over QUIC.

Go tops API clients

~12% of automated API requests
(then Node.js, Python 9.6%, Java 7.4%, .NET 3.6%).

225 Major Disruptions: Government Shutdowns Lead Outage Causes

Outage Center observations across 2024



Cause Breakdown	
Government Directed	116+
Cable Cut	41
Power Outage	16
Unknown	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Security and email threat patterns remain elevated

6.5%

of all traffic
was mitigated

Most attacked sector

Gambling /
Games

68% of global bot traffic originated
from just the top 10 countries.

Log4j remains active

CVE-2021-44228 still exploited
more than two years later.

Email threat metrics

4.3%

emails were malicious

43

of malicious emails
contained deceptive links

.bar

top-level domain generated
the largest malicious/spam share

Five Takeaways Shaping the 2025 Internet Outlook

Strategic implications for network and security leadership

1

Traffic growth remains robust: global Internet traffic increased 17% YoY, with Starlink up 3.3x.

2

Post-quantum momentum is real: TLS 1.3 PQ traffic climbed from ~2% to ~13% after Chrome 124.

3

Connectivity is fragile: 225 major disruptions, led by 116+ government-directed shutdowns.

4

Threat pressure persists: 6.5% traffic mitigated, Log4j still exploited, email abuse remains significant.

5

Developer and protocol shifts continue: HTTP/3 reached 21% and Go led API client traffic (~12%).

2025 focus: accelerate modernization while hardening resilience across routing, applications, and security operations.